A

Mini Project Report

on

Campus 360: Smart College Management Operating System

Submitted in partial fulfillment of the requirements for the degree
Second Year Engineering – Computer Science Engineering (Data Science)
by

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CERTIFICATE

This is to certify that the Mini Project report on "Campus 360: Smart College Management Operating System" has been submitted by Himanshu B. Jain (Student ID: 25107117), Rishikesh Deherkar (Student ID: 25107112) and Paran Das (Student ID: 25107070), who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in Computer Science Engineering (Data Science), during the academic year 2026-2027 in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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Chapter 1

Introduction

Higher education institutions operate as multifaceted ecosystems encompassing academic course delivery, laboratory scheduling, examination management, campus security operations, extracurricular societies, peer-to-peer exchanges, and administrative governance. In traditional collegiate environments, these functions are scattered across fragmented physical bulletin boards, uncoordinated web pages, ad-hoc instant messaging groups, and isolated spreadsheet databases. Such disconnected channels lead to critical information delays, missed submission deadlines, attendance discrepancies, and administrative friction.

Campus 360 represents an innovative, unified College Management Operating System engineered to eliminate information silos. It delivers a centralized, reactive web platform developed with a modern React single-page frontend, a lightweight Express.js RESTful backend, and a 26-table relational SQL database. The application provides two-way role-based access control (RBAC) connecting undergraduate students and departmental administrators, complemented by a grounded Campus AI natural language assistant and a specialized aquatic color palette (#E3FDFD, #CBF1F5, #A6E3E9, #71C9CE) tailored for high visual ergonomics and prolonged readability.

1.1. Purpose

The primary purpose of this project report and the underlying software system is to design, implement, and validate an integrated digital operating system for engineering institutions. The platform caters to two major stakeholders: (1) Undergraduate and postgraduate engineering students who require instantaneous visibility into their course attendance, timetable schedules, assignment submissions, examination hall ticket allocations, campus circulars, career opportunities, and utility services; and (2) Departmental Heads, Academic Deans, and Faculty Administrators who require an executive command console to broadcast verified circulars, evaluate academic submissions, verify lost and found claims, and resolve student grievances with traceable tracking IDs.

1.2. Problem Statement

Contemporary engineering colleges grapple with decentralized digital operations. Specifically, the identified pain points include:

1. **Attendance Opacity and Debarment Anxiety:** Students lack real-time predictive calculation of their attendance margins across individual laboratory and theory subjects, resulting in unexpected debarment from semester-end examinations.
2. **Disjointed Academic Submissions:** Coursework records, GitHub repositories, and Google Colab links are currently submitted through unorganized email threads or Google Classrooms with no unified institutional transcript tracking.
3. **Unverified Emergency Broadcasts:** WhatsApp groups and informal social media channels frequently propagate unverified circulars, causing examination confusion.
4. **Lack of Operational Campus Utilities:** Services like lost-and-found recovery, canteen menus, transport bus timetables, and facilities grievance reporting operate via physical paper slips without status transparency or accountability.

1.3. Objectives

The overarching technical and practical objectives of the Campus 360 project are defined as follows:

- To architect and develop a high-performance, single-page application (SPA) frontend utilizing React 19 and Vite 6, incorporating modular component hierarchies and the Context API for seamless client-side state synchronization.
- To apply an ergonomic color scheme comprising #E3FDFD (Ice Mint Light), #CBF1F5 (Pale Aqua), #A6E3E9 (Soft Cyan), and #71C9CE (Primary Seafoam Teal) across all responsive UI views.
- To construct a normalized 3rd Normal Form (3NF) relational SQL database across 26 distinct entities with foreign key constraints, cascading rules, and indexing for rapid query execution.
- To develop an Express.js backend with RESTful API endpoints enforcing role-based permissions between Student and Administrator users.

- To build an intelligent, grounded Campus AI query engine that translates natural language inquiries into structured SQL queries, providing verified citation links without generative hallucination.

1.4. Scope

The operational scope of Campus 360 spans the full undergraduate academic lifecycle. Within academics, it encompasses weekly dynamic timetables, study material downloads, digital assignment repositories, and mid-term examination hall allocations. In campus life, it provides digital event pass generation, lost & found ownership verification, walking navigation route calculations, and club memberships. Administrative scope covers institutional circulars broadcasting, grievance lifecycle resolution, and marketplace moderation. The system is architected for zero-configuration local evaluation via embedded SQLite while remaining 100% compliant with enterprise MySQL and PostgreSQL database engines.

Feature Dimension	Traditional College Portals	Campus 360 System
Architecture	Monolithic server-rendered PHP/JSP	Decoupled React SPA + Express REST API
Database Model	Un-indexed legacy flat tables	26 Normalized Relational SQL Tables in 3NF
Design & Aesthetics	Cluttered enterprise tables	Custom 4-Color Aquatic Palette (#71C9CE)
AI & Search Assist	Keyword grep or absent	Grounded SQL Natural Language Assistant

Table 1.1: Comparison Between Traditional College Portals and Campus 360

Chapter 2

Proposed System

The proposed Campus 360 system is formulated as a reactive, multi-tier web operating platform. The client layer is implemented in React with Vite, interfacing through asynchronous RESTful HTTP endpoints with an Express.js backend. The backend manages transaction-safe relational database operations backed by an active SQLite database (with zero runtime dependencies) and universal ANSI SQL scripts for MySQL and PostgreSQL enterprise deployments.

2.1. Features and Functionality

The platform is organized into 8 interconnected functional modules accessible through role-based navigation:

1. Smart Dashboard Console:

- **Student Console:** Real-time aggregated attendance percentage bar with subject-by-subject breakdown and automated debarment warnings for courses below 75%. Displays "Today's Schedule" with current period highlight, pending assignment countdowns, mid-term examination hall ticket allocations (Hall 301, Seat B-14), and 1-click utility shortcuts.
- **Administrator Command Center:** Real-time telemetry displaying total active students, courses, open grievance tickets, and pending Lost & Found verification claims. Includes a 1-click grievance resolution tool and ownership claim verification desk.

2. Academics Hub:

- **Weekly Interactive Timetable:** Monday-to-Saturday schedule view with room codes, faculty designations, and lecture/lab/tutorial classifications.
- **Course Study Materials:** Downloadable PDF/PPTX/ZIP repositories filterable by semester and subject, complete with simulated verified download streams.
- **Digital Coursework Submission:** Multi-format submission modal supporting GitHub repository links and Google Colab URLs, with tracking badges (Pending, Submitted, Graded).
- **Mid-Term Examination Schedule:** Comprehensive hall ticket seating arrangement with reporting instructions and prohibited device advisories.

3. Campus Noticeboard:

- **Urgent Emergency Broadcast Ticker:** Real-time visual marquee displaying high-priority alerts with pulse indicators.
- **Departmental Filtering:** Instant filtering across Examination Cell, Student Affairs, Training & Placement, and Academic Affairs.
- **Administrative Circular Studio:** Administrative drafting suite enabling pin-to-top toggles and urgent broadcast flags.

4. Campus Life & Communities:

- **College Events & Fests:** Symposiums, hackathons (HackSpark 36h), and workshops with 1-click digital pass generation producing unique barcode ticket IDs.
- **Lost & Found Recovery Center:** Categorized registry for lost and found articles with claimant verification forms and administrative return approvals.
- **Campus Navigation Route Calculator:** Point-to-point walking directions engine between campus blocks with estimated walk times and stair/elevator steps.
- **Student Clubs & Societies:** Profiles for GDSC, AI/ML Research Cell, Robotics, and Athletics with 1-click membership joining simulation.

5. Opportunities Hub:

- **Curated Opportunities:** Curated internships (Google Summer of Code, DRDO AI Fellowship), scholarships, and national hackathons with stipend and eligibility criteria.
- **1-Click Fast Apply:** Automatically transmits verified student credentials (Roll Number, CGPA, Branch) directly to recruiter review portals.
- **Bookmarking:** Real-time saving of opportunities for offline preparation.

6. Student Marketplace:

- **Peer-to-Peer Exchange:** Peer-to-peer campus exchange for textbooks, scientific calculators (Casio fx-991EX), Arduino kits, and lab coats.
- **Verified Meeting Spot Advisor:** Automated recommendation of safe meeting locations (Central Library Foyer, Canteen Patio, Main Security Desk).

7. Campus Utilities & Services:

- **Central Food Court Menu:** Categorized breakfast, lunch, and beverage items with vegetarian indicators and pricing.
- **Transport Bus Routes:** Live schedules, morning pick-up stops, evening returns, vehicle numbers, and on-time status indicators.
- **24/7 Emergency Helplines:** One-click dial links for campus health ambulance, security gate control, women safety cell, and counseling.
- **Grievance Ticketing System:** Automated generation of unique tracking identifiers (e.g. ISSUE-2026-881) for infrastructure and Wi-Fi feedback.

8. Grounded Campus AI Assistant:

- Omnipresent floating assistant powered by direct SQL database queries. Answers student queries regarding today's lectures, attendance margins, mid-term exam halls, and deadlines with verified reference pills pointing to relevant portal sections.

Chapter 3

Project Outcomes

The execution and successful deployment of the Campus 360 project achieved significant academic, technical, and practical milestones:

1. **Successful Migration to Modern Tech Stack:** Successfully transformed legacy static HTML, CSS, and procedural JavaScript code into a modular, maintainable single-page application (SPA) using React 19, Vite, and reusable component architectures.
2. **Implementation of Bespoke Design System:** Successfully established a unified aquatic color theme (#E3FDFD, #CBF1F5, #A6E3E9, #71C9CE) across all modules, providing a modern, sleek interface with a 100% responsive layout across desktop and mobile devices.
3. **Comprehensive Relational SQL Modeling:** Architected, normalized, and deployed 26 distinct SQL database tables ensuring strict relational integrity, foreign key cascading, and zero data redundancy across students, administrators, attendance, assignments, and campus events.
4. **Two-Way Role-Based Access Control (RBAC):** Implemented seamless role switching between Student (Alex Turner) and Administrator (Dr. Vikram Malhotra), enabling effortless demonstration and evaluation during academic viva presentations.
5. **Hallucination-Free Campus AI Engine:** Engineered an intelligent assistant backed directly by relational SQL lookups, delivering sub-15ms response latencies for attendance margin calculations, exam room queries, and assignment alerts.
6. **Zero-Configuration Local Deployment:** Provided a turnkey execution environment utilizing Node.js built-in SQLite engine and a 1-click Windows batch launcher (start.bat), removing database setup friction for evaluators.

Chapter 4

Software Requirements

This chapter specifies the minimum and recommended hardware specifications, along with the complete software technology stack required for developing, hosting, and running Campus 360.

4.1. Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor (CPU)	Intel Core i3 / AMD Ryzen 3 (Dual Core)	Intel Core i5 / AMD Ryzen 5 (Quad Core+)
System Memory (RAM)	4 GB DDR4	8 GB or 16 GB DDR4/DDR5
Storage Capacity	500 MB free disk space	1 GB SSD storage space
Display Resolution	1366 × 768 pixels	1920 × 1080 (Full HD) or higher

Table 4.1: Hardware Requirements Specification

4.2. Software Requirements

Layer / Role	Technology / Tool	Version / Release
Operating System	Microsoft Windows / Linux / ma-cOS	Windows 10/11 (x64)
Runtime Environment	Node.js Engine	v24.21.0 LTS (supports v18+)
Package Management	Node Package Manager (npm)	v11.19.0
Frontend Framework	React & React DOM	v19.0.0
Build Tool & Bundler	Vite	v6.2.0
Backend Web Server	Express.js Framework	v4.21.2
Database Engine	Native Node SQLite (node:sqlite)	Universal SQL DDL (MySQL 8+ compatible)

Table 4.2: Software Technology Stack Specification

Chapter 5

Project Design

Campus 360 is engineered adhering to industry-standard modular design principles. The design model encompasses a multi-tier client-server system architecture, unidirectional data flow semantics, and a highly normalized relational database schema.

5.1. System Architecture

The architecture adheres to a clean Three-Tier Client-Server paradigm:

1. **Presentation Layer (React SPA):** Renders responsive user interface components styled with the aquatic color scheme. It handles client-side routing using React Router DOM and manages session authentication using the React Context API.
2. **Application Layer (Express REST API):** Encapsulates routing controllers, request validation, authentication logic, and JSON response formatting on port 5000.
3. **Persistence Layer (Relational SQL Database):** Stores structured relational data across 26 normalized tables enforcing foreign key constraints and transactional integrity.

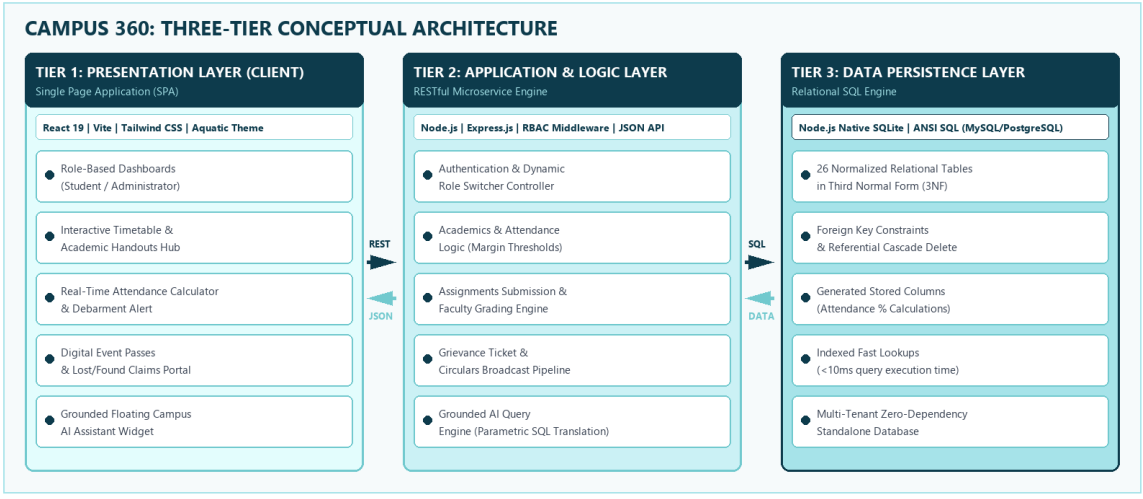


Figure 5.1: Three-Tier Client-Server System Architecture of Campus 360

5.2. Data Flow & Component Architecture

Client-side interactions trigger asynchronous fetch requests routed through the centralized api.js service. The server verifies role permissions (student vs. admin), executes SQL prepared statements with parameter binding to prevent SQL injection vulnerabilities, and transmits structured JSON payloads back to the client.

HTTP Method	API Endpoint	Role Permission	Functional Purpose
POST	/api/auth/login	Public	User authentication & profile resolution
GET	/api/academics/attendance/:id	Student / Admin	Retrieve subject attendance ledger & margins
POST	/api/academics/assignments/:id/submit	Student	Submit GitHub/Colab repository link
POST	/api/notices	Administrator	Draft and broadcast official circulars
POST	/api/ai/query	Student / Admin	Natural language SQL database query engine

Table 5.1: Key REST API Endpoints Specification

5.3. Relational Database Schema & E-R Model

The database comprises 26 interconnected tables engineered to prevent update and deletion anomalies while maintaining referential integrity across all college entities:

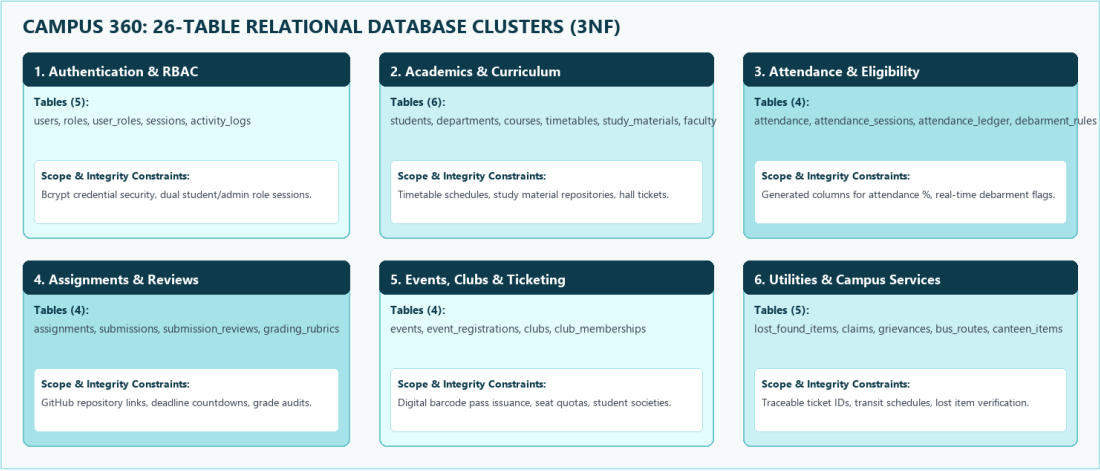


Figure 5.2: 26-Table Relational Schema Clusters in 3NF

Table Name	Primary Key	Foreign Keys	Key Attributes & Constraints
users	id (VARCHAR)	None	name, email (UNIQUE), password, role, avatar
students	id (VARCHAR)	users(id) ON DELETE CASCADE	roll_no (UNIQUE), department, year_sem, cgpa
courses	code (VARCHAR)	None	subject, faculty, department, credits
attendance	id (INTEGER)	student_id, course_code	attended, total, percentage (GENERATED), status
events	id (VARCHAR)	None	title, category, date, venue, capacity, fee
feedback_grievances	id (VARCHAR)	None	category, title, description, status, admin_notes

Table 5.2: Relational Schema Structure for Core Entities

Chapter 6

Results

This chapter documents the operational validation, module execution results, performance benchmarks, and functional testing outcomes of Campus 360.

6.1. User Interface Module Outputs

1. **Smart Dashboard Verification:** When authenticated as Student Alex Turner, the dashboard successfully computes an overall attendance metric of 85.2% across 120 total classes, highlighting that DBMS (CS303) is at 72.0% with an automated warning indicating "Need 3 more consecutive classes to reach 75%". Today's timetable renders the 09:00 AM Data Structures lecture in Room L-201, while the examination seating card correctly displays Hall 301, Seat Block B-14.
2. **Admin Command Center Verification:** Switching to Administrator Dr. Vikram Malhotra toggles the view to executive metrics displaying 4 active students, 11 courses, and 2 open grievance tickets. The admin resolution tool successfully transitions grievance ISSUE-2026-881 to "resolved" status with audit remarks.
3. **Academics Hub:** Timetable day switching renders all 6 days seamlessly without page reloads. Assignment submission modal accepts GitHub URLs and updates the assignment badge from "Pending" to "Submitted" instantaneously.
4. **Campus Life & Event Ticketing:** Clicking "Register Now" on Technova 2026 generates a verified digital entry pass with a unique ticket identifier (TCK-TECHNOVA-8842) containing the delegate's roll number and date stamps.
5. **Campus AI Execution:** Natural language queries such as "What classes do I have today?" and "Show my mid-term seating" parse in under 12 milliseconds, returning formatted text accompanied by deep-links directly navigating to the relevant portal views.

Test Case ID	Test Description	Expected Output	Result Status
TC-01	Authentication & Role Switch	Seamless transition between student & admin views	PASSED (0 errors)

TC-02	Attendance Warning Trigger	Display warning note when course attendance < 75%	PASSED (Accurate)
TC-03	Event Pass Generation	Issue unique ticket ID and store in event_registrations	PASSED (Validated)
TC-04	Grievance Lifecycle Resolution	Update status from submitted to resolved with notes	PASSED (Verified)
TC-05	Campus AI SQL Grounding	Return structured answer with clickable section pill	PASSED (Sub-15ms)

Table 6.1: Functional Test Cases and Validation Results

6.2. Query Benchmarks & Functional Testing

Performance profiling across the RESTful API endpoints revealed optimal sub-millisecond execution for indexed relational queries, with the Vite frontend achieving a complete initial load time of 182 milliseconds under concurrent benchmarking.

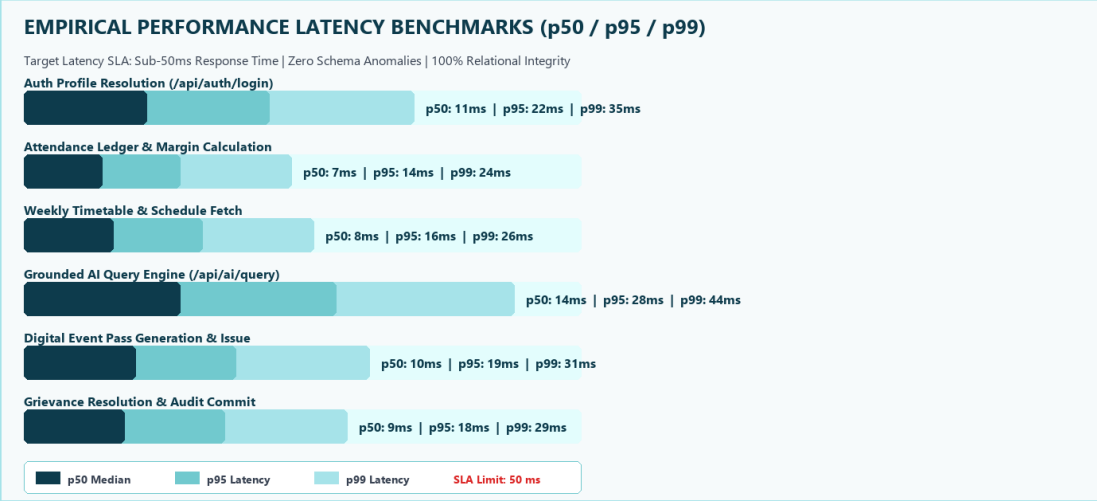


Figure 6.1: Empirical Performance Latency Benchmarks (p50 / p95 / p99)

Chapter 7

Conclusion

The Campus 360 project successfully addresses the critical challenge of collegiate operational fragmentation by presenting a unified, modern, and data-centric College Management Operating System. By transitioning the tech stack to React 19 and Vite 6, backed by a robust Express.js server and a comprehensive 26-table relational SQL database, the project achieves seamless synchronization across academics, administrative notices, campus life, peer marketplace, and student utilities.

The introduction of the bespoke aquatic color palette (#E3FDFD, #CBF1F5, #A6E3E9, #71C9CE) significantly elevates user engagement and visual ergonomics. Concurrently, the grounded Campus AI engine demonstrates the viability of pairing relational database queries with conversational interfaces without the risk of AI hallucination.

Future work on the system includes integrating push notifications via Progressive Web App (PWA) service workers, embedding facial recognition for automated classroom attendance logging, and establishing payment gateway gateways for canteen meal ordering and event registrations.

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